

2520090008-Sathish kumar

Task-1:

To Create Main Loop, Display Prompt, Read User Input, Handle Exit Conditions, Design Control Flow Diagram, Test Interactive Loop

```
root@LAPTOP-4H877K4U:~# nano task1.c
```

```
#include <stdio.h>
#include <string.h>
int main()
{
    char command[100];
    while(1)
    {
        printf("Enter a command (type exit to quit): ");
        fgets(command, sizeof(command), stdin);
        command[strcspn(command, "\n")] = 0;
        if(strcmp(command, "exit") == 0)
        {
            printf("Exiting Program...\n");
            break;
        }
        printf("You entered: %s\n", command);
    }
    return 0;
```

Then press ctrl+o enter ctrl+x2) then gcc filename.c -o filename

```
root@LAPTOP-4H877K4U:~# gcc task1.c -o task1
```

```
root@LAPTOP-4H877K4U:~# ./task1
```

enter the linux command

```
root@LAPTOP-4H877K4U:~# nano task1.c
```

```
root@LAPTOP-4H877K4U:~# gcc task1.c -o task1
root@LAPTOP-4H877K4U:~# ./task1
Enter a command (type exit to quit): hello
You entered: hello
Enter a command (type exit to quit): exit
Exiting Program...
root@LAPTOP-4H877K4U:~#
```

Design Control Flow Diagram

```
Start
|
Display Prompt
|
Read User Input
|
Is Input "exit"?
|  |
No  Yes
|  |
Display  Exit Program
Input    |
|        |
|        |
<-----
```

Task-2

Step 1: Create File1

```
nano File1.txt
```

Write:

Welcome to Ubuntu Lab

Step 2: Create File2

```
root@LAPTOP-4H877K4U:~# nano file1.txt
root@LAPTOP-4H877K4U:~# touch File2.txt
root@LAPTOP-4H877K4U:~#
```

Step 3: Create C Program

```
root@LAPTOP-4H877K4U:~# nano file1.txt
root@LAPTOP-4H877K4U:~# touch File2.txt
root@LAPTOP-4H877K4U:~# nano copyfile.c
```

```
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>
int main()
{
    char buffer[100];
    int source, destination, bytes;
    // open() - Opens File1 for reading
    source = open("File1.txt", O_RDONLY);
    // open() - Opens File2 for writing
    destination = open("File2.txt", O_WRONLY | O_CREAT | O_TRUNC, 0644);
    // read() and write() are used inside the loop
    while((bytes = read(source, buffer, sizeof(buffer))) > 0)
    {
        write(destination, buffer, bytes);
    }
    // close() - Closes both files
    close(source);
    close(destination);
    printf("File copied successfully.\n");
    return 0;
}
```

Compile and Run

```
root@LAPTOP-4H877K4U:~# nano file1.txt
root@LAPTOP-4H877K4U:~# touch File2.txt
root@LAPTOP-4H877K4U:~# nano copyfile.c
root@LAPTOP-4H877K4U:~# gcc copyfile.c -o copyfile
root@LAPTOP-4H877K4U:~# ./copyfile
```

OUTPUT

```
root@LAPTOP-4H877K4U:~# nano file1.txt
root@LAPTOP-4H877K4U:~# touch File2.txt
root@LAPTOP-4H877K4U:~# nano copyfile.c
root@LAPTOP-4H877K4U:~# gcc copyfile.c -o copyfile
root@LAPTOP-4H877K4U:~# ./copyfile
File copied successfully.
root@LAPTOP-4H877K4U:~#
```